Adaptive Tutoring Beyond Classical RAG: Conditional Retrieval in a Repository-Grounded Multi-Agent Study

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***Abstract*— Retrieval-augmented generation grounds tutor responses in external evidence, but effective tutoring also requires learner diagnosis, scaffold sequencing, and rubric alignment. We study adaptive tutoring as an *architecture-selection* problem: should retrieval control every interaction, or should it activate only when explicit evidence is needed? Under shared repository-grounded controls, we compare four configurations—classical RAG, non-RAG multi-agent coordination, a single-agent baseline, and a conditional-retrieval hybrid—built on the same Qwen3.5-4B backbone (thinking budget 512). On TutorEval [1] ($n = 834$, identical examples), the hybrid achieves the best composite score (0.383), best Token-F1 (0.115), best tutor-hit rate (0.957), and lower latency/tokens than classical RAG (-5.4% latency, -12.2% tokens). On the latest synchronized EduBench [2] snapshot, captured once every architecture had processed at least 1,580 examples, hybrid and non-RAG multi-agent tie on composite (0.227) and EB12D mean (0.398), with hybrid slightly better rubric coverage and latency. The results support a systems view in which retrieval is a conditional capability inside a supervisory tutoring policy rather than the universal top-level controller.**

I. Introduction

A student asks a tutoring system, "What does this concept mean?" The right answer is a short explanation grounded in source material—essentially a retrieval problem. Another student asks, "Why is my answer wrong?" That task is different: the tutor must decide whether to explain directly, offer a scaffolded hint, align feedback to a rubric, or deliberately withhold part of the answer to preserve the learning opportunity. What looks like a retrieval problem has become an instructional decision problem.

This paper studies adaptive tutoring through that lens. We frame tutoring as a *regime-dependent architecture-selection problem* and ask a focused systems question: should retrieval control every interaction, or should it activate only when explicit evidence is needed? To answer, we build a repository-grounded multi-agent framework and compare four configurations—classical RAG, non-RAG multi-agent coordination, a single-agent baseline, and a proposed hybrid with *conditional* retrieval—under shared repository-grounded controls.

Retrieval-augmented generation (RAG) grounds model outputs in external evidence [3] and remains a strong default for fact-seeking question answering. Tutoring, however, requires more than grounded recall. A useful tutor must infer learner state, choose how much help to provide, follow pedagogical constraints, and decide when instructional strategy should take priority over direct answer delivery. The question is therefore not whether retrieval helps, but *when* retrieval is sufficient and when additional tutoring control matters.

Recent education-centered benchmarks make that distinction harder to ignore. EduBench and TutorEval evaluate not only correctness but pedagogy, adaptivity, reasoning, and grounded tutoring behavior [1], [2], [4], and a growing evaluation literature emphasizes open-ended pedagogical quality, adaptivity, and safety over answer accuracy alone [5]–[10]. In these settings, a system can be factually correct yet still tutor poorly by giving the wrong level of help, missing rubric criteria, or failing to adapt.

We use *orchestration* to denote the additional control required for such cases: diagnosis, planning, routing, rubric use, critique, and intervention *before or during* answer generation. Our framework makes this control explicit through regime-aware routing and conditional retrieval. From a systems-and-cybernetics perspective, the router acts as a lightweight supervisory controller over retrieval and pedagogical sub-policies, deciding when evidence grounding should dominate and when instructional coordination should dominate. Rather than assuming one architecture governs every interaction, we compare the four configurations above under identical infrastructure. Retrieval is available when evidence is needed, but it is not treated as the controller by default.

Contributions. (i) We formalize adaptive tutoring as architecture selection conditioned on a lightweight regime router and operationalize it in a repository-grounded framework that instantiates classical RAG, non-RAG multi-agent, single-agent, and hybrid conditional-retrieval pipelines on a shared substrate, with the router serving as a supervisory controller over retrieval and pedagogical sub-policies. (ii) We report complete TutorEval results on identical examples ($n = 834$) and an updated synchronized EduBench snapshot anchored at a common processed floor of 1,580 examples per architecture. (iii) We show that the proposed hybrid delivers the strongest overall quality-efficiency balance without treating retrieval as the default top-level controller.

II. Related Work

Our work sits at the intersection of three lines: educational tutoring benchmarks, retrieval control for RAG, and multi-agent orchestration. We highlight where each line stops and why the boundary between them motivates the experiments in this paper.

Educational tutoring benchmarks. Recent educational benchmarks move the evaluation target beyond answer correctness toward pedagogy, adaptivity, and instructional

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safety. EduBench spans nine tutoring scenarios and scores educational utility dimensions such as adaptation and pedagogical application [2]. TutorEval operationalizes chapter-grounded science tutoring with keypoint-based evaluation [1], [4]. MathTutorBench, TutorBench, and related evaluation frameworks sharpen open-ended pedagogical assessment and tutor-specific safety [5]–[10]. These benchmarks crystallize the gap our paper targets: a tutor can be factually correct yet still tutor poorly by giving the wrong level of help, neglecting rubric criteria, or responding unsafely, and optimizing for answer correctness alone will not recover that behavior.

RAG and agentic retrieval control. Classical RAG conditions generation on retrieved passages [3] and is a strong default for fact-seeking QA. Subsequent work shows that retrieval quality depends on the *control policy* around it: Self-RAG trains reflection tokens that decide when to retrieve and when to critique an answer [11]; CRAG adds a lightweight retrieval evaluator that triggers corrective actions when retrieved evidence is poor [12]; and adaptive RAG variants have been studied for short-form open-domain QA [13]. A parallel evaluation literature diagnoses retrieval-heavy systems through answer quality, citation behavior, and benchmarking infrastructure [14]–[17]. In tutoring, however, many queries are dominated by diagnosis, hinting, or rubric-following rather than evidence-seeking, and long educational contexts can make naive "retrieve by default" behavior brittle [18], [19]. Unlike Self-RAG's learned reflection tokens or CRAG's retrieval evaluator, our router is a training-free, *sub-model-call* heuristic over lexical cues and dataset priors (§III), which keeps latency and token costs predictable in resource-constrained tutoring deployments—a positioning orthogonal to, rather than competing with, learned adaptive RAG.

Multi-agent tutoring and coordination-heavy control. A broad multi-agent literature argues that explicit role specialization helps with planning, critique, and decomposition in complex workflows [20]–[23]. In retrieval settings, collaborative agents have also been proposed to improve search and reasoning over evidence [24], [25]. Closer to our problem, recent tutoring systems including AgentTutor, IntelliCode, and CogEvo-Edu instantiate specialized educational roles such as learner modeling, multi-turn teaching, and collaborative tutoring [26]–[28]. These designs motivate tutoring-specific orchestration, but they do not isolate the architectural question we study: *when* should evidence retrieval be the primary controller, and when should it remain a subordinate capability inside a broader tutoring policy?

Positioning. Prior work supplies the ingredients—educational benchmarks that reward pedagogy, retrieval methods that improve grounding, and multi-agent frameworks that improve coordination—but leaves the boundary between them under-specified. Our contribution is a repository-grounded comparison that makes this boundary explicit by testing retrieval-first, coordination-heavy, and hybrid conditional-retrieval control under shared infrastructure and freeze-time conditions.

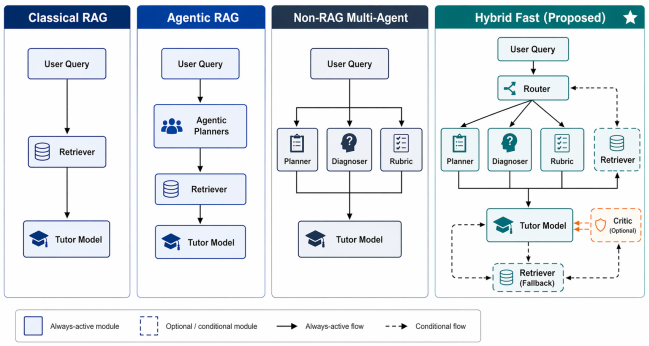

Fig. 1. Repository-grounded execution families used in the comparison. The proposed `hybrid_fast` path combines router-mediated specialist coordination with conditional retrieval, optional critic revision, and a single fallback retrieval pass.

III. Methodology

We now describe the framework and the experimental protocol used for the frozen comparison.

The framework implements five execution families—classical RAG, agentic RAG, non-RAG multi-agent, single-agent baseline, and our proposed conditional-retrieval hybrid (`hybrid_fast`). A lightweight router selects the family and sets execution flags for each example (Fig. 1). The paper comparison uses four families with comparable outputs; the implemented `agentic_rag` path is excluded because parity-controlled outputs were unavailable. The design principle is *controlled adaptivity*: invoke retrieval only when an instance needs external evidence, and otherwise spend the saved budget on pedagogical coordination.

A. Regime-aware routing

For each example x, the router computes five heuristic scores $s(x) \in [0,1]^5$ from normalized keyword cues (evidence, coordination, rubric, planning, tutoring), augmented with dataset priors (e.g., TutorEval→*adaptive tutoring*, HotpotQA→*evidence grounded*). An optional logistic-regression classifier can override the predicted family when a trained router is loaded; the frozen runs use the heuristic+prior fallback throughout. The selected family and the retrieval gate for `hybrid_fast` follow:

$$\text{family}(x) = \begin{cases} \text{CLS-RAG} & s_e(x) \geq \tau_e^{\text{cls}},\ s_c(x) < 0.35 \\ \text{AGN-RAG} & s_e(x) \geq 0.35,\ s_c(x) \geq \tau_c \\ \text{NON-RAG} & s_c(x) \geq 0.55,\ s_e(x) < 0.28 \\ \text{HYB-FAST} & \text{otherwise}, \end{cases} \tag{1}$$

where s_e, s_c are the evidence and coordination scores and $(\tau_e^{\text{cls}}, \tau_c) = (0.52, 0.50)$. Within `hybrid_fast`, the retrieval gate is

$$\text{require_retrieval}(x) = \mathbb{1}[s_e(x) \geq 0.35 \ \vee \ \text{regime}(x) = \text{EG}], \tag{2}$$

where EG denotes the *evidence-grounded* regime. A fallback retrieval call is issued when the primary path returned *no* chunks but $s_e(x) \geq 0.45$ (a small safety margin above the gate), and the tutor is re-invoked with the recovered evidence. These thresholds are the production values shipped with the repository; we report sensitivity as a planned ablation in §IV.

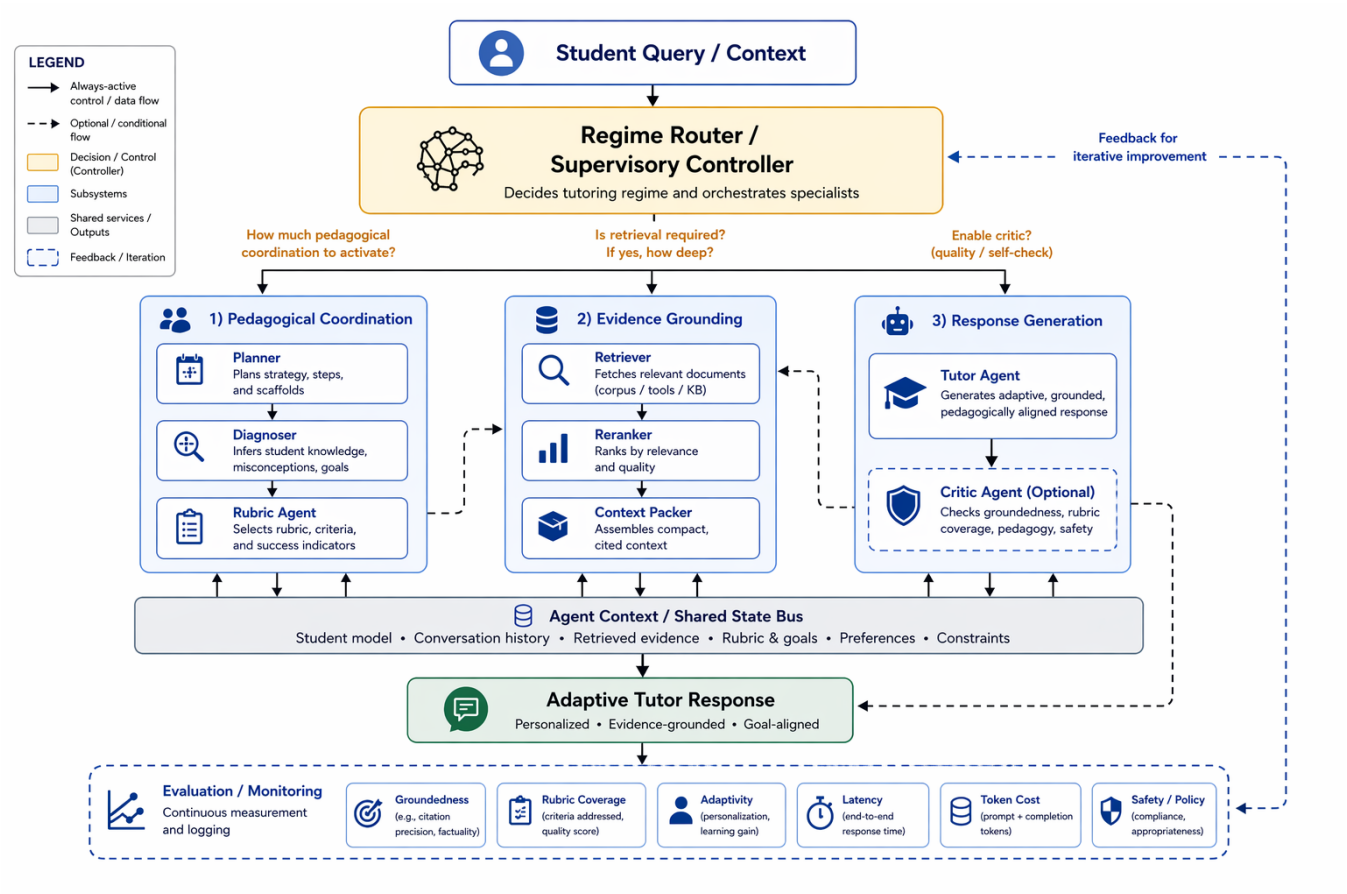

Fig. 2. Full system view of the proposed adaptive tutoring controller. The regime router acts as a supervisory controller that allocates pedagogical coordination, evidence grounding, and response generation through a shared state bus and a monitoring loop.

B. Specialist agents and shared context

All execution families operate on a shared `AgentContext` carrying the example, route decision, student state, search queries, rubric summary, retrieved chunks, and draft answer. Specialists are lightweight and launched in parallel where possible: the **planner** emits up to three search queries tailored to the regime; the **diagnoser** infers learner level, goals, style, and likely misconceptions; the **rubric agent** compiles a checklist from rubric items; the **tutor** drafts the final response using retrieved evidence when available; and an optional **critic** inspects the draft for concrete issues (missing evidence markers, unmet rubric items, length mismatches) and revises only when such issues are detected. All agents share the same small model backbone and shared budget policy, so differences across families reflect *structural* rather than capacity differences.

C. Hybrid retrieval, reranking, and context packing

The retrieval stack is intentionally lightweight. A hybrid index combines word-level TF–IDF ($\lambda_w = 0.58$), character n-gram TF–IDF ($\lambda_c = 0.17$), and an optional truncated-SVD latent projection ($\lambda_\ell = 0.25$); the weighted similarity is boosted by query–title token overlap. A six-feature logistic reranker (base score, token/title overlap, phrase match, numeric overlap, brevity) reorders the candidates, and a maximum marginal relevance [29] packer with $\lambda_{\text{MMR}} = 0.68$ selects up to four chunks within a 6,000-character context budget.

D. Evaluation protocol

Metric handling is *profile-aware* and enforced in code. Shared metrics include composite score, token-level F1, rubric coverage, latency, token cost, and retrieval-sensitive grounding signals; EduBench additionally reports its 12-dimension educational rubric and score alignment, while TutorEval reports tutor-hit and chapter-grounded tutoring quality. When a metric is profile-inapplicable, we omit it from the headline comparison rather than forcing structural zeros into the main table.

IV. Experimental Results

This section reports the updated paper numbers: finished TutorEval results and the latest synchronized EduBench dashboard snapshot. We emphasize answer quality and efficiency rather than run-completion accounting.

A. Setup

Compared methods. The comparison uses four configurations (Fig. 1): (a) classical RAG (retrieval followed by tutor); (b) non-RAG multi-agent (planner + diagnoser + rubric specialists, no retrieval); (c) single-agent (tutor only); and (d) the proposed `hybrid_fast` (Fig. 2) with parallel specialist preparation, conditional retrieval, and fallback. The

implemented agentic_rag path is not included because parity-controlled outputs were unavailable.

Model and hyperparameters. All architectures use the same open-weight backbone (QWEN3.5-4B) with a fixed reasoning/"thinking" budget of 512 tokens via an OpenAI-compatible endpoint, temperature 0.15, and a per-response cap of 900 tokens. Retrieval settings are unchanged from §III: word/character/latent weights $(0.58, 0.17, 0.25)$, latent dim 128, top-$k = 8$, final-$k = 4$, chunk size 220 with 40-token overlap, and MMR $\lambda = 0.68$. Budget caps remain 3 retrieval queries, 4 tool calls, 4 agents, one critic round, and a $6{,}000$-character context window.

Benchmarks and reporting policy. EduBench spans nine educational scenarios and twelve teacher- and student-facing aspects [2]; TutorEval assesses chapter-grounded science tutoring with keypoint-oriented evaluation [1]. All systems share the same indexed corpus, reranker, packer, prompt scaffold, and Qwen3.5-4B backbone. TutorEval is complete ($n = 834$ across all four architectures on identical examples). EduBench is summarized from the latest synchronized snapshot, captured on 2026-04-22 once each architecture had processed at least 1,580 examples; we use this common processed floor as the reporting anchor and omit success-rate tables.

B. Updated quality-efficiency comparison

Table I summarizes the updated comparison. On TutorEval ($n = 834$, identical examples), hybrid_fast attains the strongest overall balance: best composite (0.383), best Token-F1 (0.115), best tutor-hit rate (0.957), best rubric coverage (0.944), and lower mean latency/tokens than classical RAG (49,113 ms vs. 51,903 ms; 3,490 vs. 3,975 tokens). Non-RAG and single-agent baselines remain competitive on some tutoring metrics but lose clearly on the composite score and efficiency-quality balance.

On the synchronized EduBench snapshot, hybrid_fast and non_rag_multi_agent are effectively tied on composite (0.227) and EB12D mean (0.398). The difference between them is operational rather than dramatic: hybrid is slightly faster (13,292 ms vs. 13,725 ms), slightly leaner in tokens (2,218 vs. 2,223), and slightly higher on rubric coverage (0.895 vs. 0.891), whereas non-RAG holds a small EduAlign edge (0.130 vs. 0.126). Classical RAG trails both coordination-heavy routes on composite and rubric despite the strongest EduAlign (0.166), and the single-agent baseline is consistently behind on composite. The updated picture is therefore not that retrieval is useless, but that retrieval-first control is not the most reliable default once pedagogical coordination becomes central.

V. Discussion

Three observations structure the take-away. *First*, the TutorEval endpoint favors the hybrid most clearly: it leads on composite, Token-F1, tutor-hit rate, and rubric coverage while also reducing latency and tokens relative to classical RAG. *Second*, the synchronized EduBench snapshot shifts the story from retrieval alone to pedagogical coordination: hybrid and non-RAG multi-agent tie on the headline educational scores, while hybrid retains a modest speed and rubric edge. *Third*, retrieval-first control is not the strongest default policy: classical RAG remains competitive on EduAlign, but it is slower and materially weaker on composite and rubric than the best coordination-heavy routes.

These patterns argue for treating adaptive tutoring as an architecture-selection problem rather than a retrieval-parameter problem. Viewed through an SMC lens, the central design object is a supervisory policy that coordinates retrieval, pedagogical planning, diagnosis, and revision under uncertainty and resource limits. Conditional retrieval is one answer; richer controllers with calibrated uncertainty, explicit per-example budgets, and human override hooks are a natural next step.

VI. Limitations

We emphasize three constraints. **(i) Snapshot EduBench reporting.** TutorEval is fully complete, but EduBench is summarized from a synchronized live snapshot captured when the slowest architecture reached 1,580 processed examples, not from a matched-slice or fully converged rerun. **(ii) No statistical testing.** We report point estimates only; paired bootstrap confidence intervals, permutation tests, and human evaluation are still needed before promoting small TutorEval deltas to strong significance claims. **(iii) Scope.** The study uses one small backbone (Qwen3.5-4B, thinking budget 512), one shared corpus, and English-language benchmarks, and it evaluates benchmark outputs rather than downstream learning gains.

VII. Conclusion

We compared retrieval-first, coordination-heavy, and hybrid conditional-retrieval designs under shared repository-grounded controls on a single shared small-model backbone. On TutorEval ($n = 834$, identical examples), hybrid_fast reaches the highest composite score (0.383), the best Token-F1/TE-hit/rubric combination, and lower latency/tokens than classical RAG. On the synchronized EduBench snapshot, hybrid and non-RAG multi-agent tie on composite and EB12D mean, with hybrid retaining a small cost and rubric edge. Taken together, the evidence supports framing adaptive tutoring as a *supervisory control problem*: retrieval should remain available, but as a selectively activated capability inside a broader pedagogical policy rather than the universal top-level controller.

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TABLE I

UPDATED QUALITY-EFFICIENCY SUMMARY. TUTOREVAL ROWS ARE FINISHED ON IDENTICAL EXAMPLES ($n = 834$). EDUBENCH ROWS ARE TAKEN FROM THE LATEST SYNCHRONIZED SNAPSHOT, CAPTURED ONCE EVERY ARCHITECTURE HAD PROCESSED AT LEAST $n_{\min} = 1580$ EXAMPLES. "–" DENOTES PROFILE-INAPPLICABLE METRICS. HIGHER IS BETTER EXCEPT LATENCY AND TOKENS.

Architecture	Comp.	Tok-F1	EB12D	TE Hit	Rubric	EduAlign	Lat. (ms)	Tokens
TutorEval (finished, $n = 834$)								
classical_rag	0.377	0.112	–	0.938	0.919	–	51903.1	3974.7
hybrid_fast	**0.383**	**0.115**	–	**0.957**	**0.944**	–	**49113.4**	**3490.3**
non_rag_multi_agent	0.371	0.090	–	0.932	0.907	–	51048.0	5959.3
single_agent_no_rag	0.360	0.086	–	0.900	0.867	–	55257.3	6466.2
EduBench (latest synchronized snapshot, common processed floor $n_{\min} = 1580$)								
classical_rag	0.219	–	0.366	–	0.706	**0.166**	19798.8	4021.2
hybrid_fast	**0.227**	–	**0.398**	–	**0.895**	0.126	**13292.3**	**2218.5**
non_rag_multi_agent	**0.227**	–	**0.398**	–	0.891	0.130	13725.0	2223.1
single_agent_no_rag	0.212	–	0.377	–	0.709	0.153	19050.4	2953.3

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